HKUST – Department of Computer Science and Engineering
COMP 2711: Discrete Math Tools for Computer Science

Spring 2024 Midterm Examination

Date: 22 March 2024 Time: 19:00–21:30

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This is a closed book closed notes exam. It consists of **16** pages and **12** questions.
- Please write your name, student ID, email, lecture section in the space provided at the top of this page.
- Please sign the honor code statement on page 2.
- Answer all the questions within the space provided on the examination paper. You may use the back of the pages for rough work. Each question is on a separate page. This is for clarity and is not meant to imply that each question requires a full-page answer. Some can be answered using only a few lines.
- Calculators are not allowed for the exam.
- You may use any facts taught in class, unless explicitly forbidden from doing so.

Problems	1	2	3	4	5	6	7	8	9	10	11	12	Total
Points	5	5	4	6	5	10	10	10	10	10	15	10	100
Score													

As part of HKUST's introduction of an honor code, the HKUST Senate has recommended that all students be asked to sign a brief declaration printed on examination answer books that their answers are their own work, and that they are aware of the regulations relating to academic integrity. Following this, please read and sign the declaration below.

I declare that the answers submitted for this examination are my own work.

I understand that sanctions will be imposed, if I am found to have violated the University regulations governing academic integrity.

Student's Name: _____

Student's Signature: _____

Problem 1. [5 points]

Arrange the following functions in order of increasing asymptotic complexity. Note that you must write function $f(n)$ before function $g(n)$ if $f(n) = O(g(n))$. No explanation is needed.

$$n,\ n\log n,\ 3^n,\ n!,\ n^2,\ 2^n,\ n^{3/2},\ \log n$$

Answer. $\log n, n, n\log n, n^{\frac{3}{2}}, n^2, 2^n, 3^n, n!$

Grading.

- 5 pts: exactly right answer.
- 4 pts: 1 error for wrong order.
- 3 pts: right grouping (there are 3 groups: logarithmic in n, polynomial in n, and exponential in n) and 2 errors.
- 1 pt: something right, e.g., $\log n$ is the first item.

Problem 2. [5 points]

Assume p, q and r are logical propositions. Which of the following statements are tautologies (T), which are contradictions (F), and which are neither of them (N)? No explanation is needed.

(a) $(p \vee q) \to (p \wedge q)$

(b) $(p \vee \neg q) \wedge (\neg p \wedge q)$

(c) $((p \vee q) \wedge r) \leftrightarrow ((p \wedge r) \vee (q \wedge r))$

(d) $(p \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge \neg(p \leftrightarrow r)$

(e) $((p \to (q \vee r)) \leftrightarrow ((p \to q) \vee (p \to r))$

Answer. (a) N (b) F (c) T (d) F (e) T

Grading.

- 1 pt for each right answer.

Problem 3. [4 points]

Let $P(x, y)$ be a propositional function. For each part below, state whether or not it is a tautology. No explanation is needed.

(a) $\exists x \forall y P(x, y) \to \forall y \exists x P(x, y)$.

(b) $\forall y \exists x P(x, y) \to \exists x \forall y P(x, y)$.

(c) $\exists x \forall y P(x, y) \to \forall x \exists y P(x, y)$.

(d) $\forall x \exists y P(x, y) \to \exists x \forall y P(x, y)$.

Answer.

(a) Tautology
(b) Not a tautology
(c) Not a tautology
(d) Not a tautology

Grading.

- 1 pt for each right answer.

Problem 4. [6 points]

(a) Consider the domain to be the set of all integers. Express the following statement using quantifiers:

"Every integer multiple of 4 can be expressed as the difference of two perfect squares."

You are only allowed to use $\forall, \exists, \wedge, \vee, \neg, \rightarrow, \leftrightarrow$, equality ($=$), power (e.g. (a^b)), addition (+,-) , multiplication($\times$) and divisibility ($|$).

(b) Express the negation of your statement in (a), such that the symbol $\neg$ does not appear in your statement. (As an example if you want to say $\neg(a = b)$ you must use $a \neq b$, and if you want to say $\neg(a \mid b)$ you must use $a \nmid b$)

Answer.

(a) possible solution 1: $\forall x \exists a \exists b\ (4x = a^2 - b^2)$;
possible solution 2: $\forall x\ (4 \mid x \rightarrow \exists a \exists b\ x = a^2 - b^2)$
(b) possible solution 1: $\exists x \forall a \forall b\ (4x \neq a^2 - b^2)$;
possible solution 2: $\exists x\ (4 \mid x \wedge \forall a \forall b\ (x \neq a^2 - b^2))$

Grading.

- 6 pts: exactly right answer.
- 5 pts: right transformation for $\exists$, $\forall$ and $\rightarrow$ but some minor problem for statements; right answer for question (a) but something wrong for (b), e.g. wrong transformation for $\rightarrow$.
- 3 pts: right transformation for $\exists$, $\forall$ and $\rightarrow$ but do not give right statements.
- 1 pts: something right, e.g. transform $\forall x$ in (a) to $\exists x$ in (b).

Problem 5. [5 points]

Let p, q, r, s, t, u denote five propositions. You are given the following set of premises:

1. $(\neg p \vee q) \rightarrow r$
2. $r \rightarrow (s \vee t)$
3. $\neg s \wedge \neg u$
4. $\neg u \rightarrow \neg t$

Using rules of inference for propositional logic, show that the truth of the above premises implies that the conclusion p is true. You do not need to write the names of the rules you are using; instead, you can just number each statement and write down from which statement(s) a new statement is derived.

Answer.

5. from 1 $\neg(\neg p \vee q) \vee r$
6. from 5 $(p \wedge \neg q) \vee r$
7. from 2 $\neg r \vee s \vee t$
8. from 3 $\neg s$
9. from 3 $\neg u$
10. from 7 and 8 $\neg r \vee t$
11. from 4 and 9 $\neg t$
12. from 10 and 11 $\neg r$
13. from 6 and 12 $p \wedge \neg q$
14. from 13 p

Grading.

- if everything is correct, 5 points
- if any step is wrong during the derivation or fail to achieve the final answer, each correct step will earn one point (maximum 3 points)

Problem 6. [10 points]

Prove that $\sqrt{2} + \sqrt{3}$ is irrational.

Answer.

For the sake of contradiction, suppose that $\sqrt{2} + \sqrt{3}$ is rational, that is, $\sqrt{2} + \sqrt{3} = a/b$, where a and b are both integers. Squaring both sides, we obtain $5 + 2\sqrt{6} = a^2/b^2$, which implies that

$$\sqrt{6} = \frac{1}{2}\left(\frac{a^2}{b^2} - 5\right) = \frac{a^2 - 5b^2}{2b^2}.$$

As $a^2 - 5b^2$ and $2b^2$ are both integers, it follows that $\sqrt{6}$ is rational.

Thus, there exist integers p and q such that $\sqrt{6} = p/q$ and $\gcd(p, q) = 1$. Hence $6 = p^2/q^2$. It follows that $p^2 = 6q^2$, which implies that p^2 is even. It follows that p is even, that is, there exists an integer k such that $p = 2k$. Substituting into the equation $p^2 = 6q^2$, we obtain $4k^2 = 6q^2$. This yields $2k^2 = 3q^2$. It follows that q^2 is even, which implies that q is even. We have shown that both p and q are even, which contradicts our assumption that $\gcd(p, q) = 1$. This proves that $\sqrt{2} + \sqrt{3}$ is irrational.

Grading.

- 5 points for converting the problem to proving $\sqrt{6}$ is irrational.
- 5 points for proving that $\sqrt{6}$ is irrational.

Problem 7. [10 points]

Let k be an even integer. Prove that it is not possible to write 1 as the sum of the reciprocals of k odd integers.

Note: As an example, if $k = 4$, then the k odd integers selected could be 3, 3, 5, and 7. Note that, in this case, the sum of the reciprocals of these k odd integers is given by

$$\frac{1}{3} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7},$$

which is not equal to 1. In this problem, you need to show that no matter how we pick the k odd integers, where k is an arbitrary even integer, the sum of the reciprocals of these k odd integers cannot equal 1.

Hint: Try to prove this by contradiction.

Answer.

Fix an even integer k. Suppose for the sake of contradiction that there exist k odd integers $a_1, a_2, \ldots, a_k$ where

$$\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_k} = 1.$$

Fractions seem to complicate the problem, so let's eliminate them by multiplying both sides by the product $a_1 \cdot a_2 \cdots a_k$:

$$\frac{a_1 \cdots a_k}{a_1} + \frac{a_1 \cdots a_k}{a_2} + \cdots + \frac{a_1 \cdots a_k}{a_k} = a_1 \cdot a_2 \cdots a_k. \qquad (*)$$

Note that every term on the LHS is an integer because the denominator is always present in the numerator, and is therefore cancelled out. We can use the fact that k is even and that each a_i is odd to observe the following:

(a) The RHS of (**??**) is the product of odd numbers, which must be odd.

(b.1) Each term in the LHS is of the form

$$\frac{a_1 \cdots a_k}{a_i} = \frac{a_1 \cdots \cancel{a_i} \cdots a_k}{\cancel{a_i}} = a_1 \cdots a_{i-1} \cdot a_{i+1} \cdots a_k,$$

which is the product of odd integers, and hence must be odd.

(b.2) The LHS is the sum of k terms, each of which is odd by the previous observation. Since k is even, that sum must be even.

Observation (a) tells us that the RHS of (**??**) is odd, but observation (b.2) tells us that the LHS is even. This contradicts the fact that LHS and RHS are equal. Therefore, we conclude that our hypothesis was wrong and that such k integers $a_1, \ldots, a_k$ do not exist.

Grading.

- 4 points for multiplying by $a_1 \cdots a_k$.
- 4 points for observations (a), (b.1), and (b.2). 1 mark is deducted if a solution does not find all of them.
- 2 points for using the observations to arrive at a contradiction.

Problem 8. [10 points]

The professor gave a mathematics exam to five students. The marks that the five students received (listed in increasing order) were 71, 76, 80, 82, and 91. The professor entered the scores in random order into a spreadsheet, which recalculated the class average after each score was entered. The professor noticed that after each score was entered, the average was always an integer. Based on this information, you need to determine the last score entered by the professor. Justify your answer fully.

Answer.

We assign variables $a = 71, b = 76, c = 80, d = 82, e = 91$. Let x_i denote the i-th score entered into the spreadsheet, for $1 \leq i \leq 5$. We know that $x_1 + x_2 \equiv 0 \pmod 2$, $x_1 + x_2 + x_3 \equiv 0 \pmod 3$, and $x_1 + x_2 + x_3 + x_4 \equiv 0 \pmod 4$. There are several approaches to find the last number.

Solution 1: First, observe that there are two odd numbers and three even numbers. Since the sum of the first 4 numbers is divisible by 4, it follows that both odd numbers must exist among the first 4 numbers. The sum of these two odd numbers modulo 4 is 2. So the other two even numbers cannot be 76 and 80 (since these are both 0 modulo 4). It follows that the last number has to be either 76 or 80. Now notice that the first two numbers either have to be both odd or both even. But they cannot be both odd because in that case, the sum of first three numbers cannot be divisible by 3 (since the sum of two available odd numbers is divisible by 3 and none of the even numbers is a multiple of 3). So the first two numbers have to be even such that 82 is among them. So in the first three numbers, one is 82, another is an odd number and another is an even number such that each has a remainder 1 mod 3. So the only options will be 76 and 91. So the last number x_5 has to be 80.

Solution 2: There exists an alternative argument that is simpler. The sequence of numbers $a, , b, c, d, e$ modulo 3 is $2, 1, 2, 1, 1$. Thus, there is only way to choose 3 numbers out of this set whose sum is $0 \pmod 3$, namely, by choosing the three numbers b, d, e. Next observe that this same sequence of numbers modulo 4 is $3, 0, 0, 2, 3$. Note that $b + d + e \pmod 4$ is $5 \equiv 1$. Since the sum of the first 4 numbers is $0 \pmod 4$, it follows that the 4th number selected must be a (since $a \equiv 3 \pmod 4$). Thus, the last number selected must be c, namely 80.

Grading.

- 2 points for the final answer.
- 4 points for arguments about the first three values.
- 4 points for argument about the first four values.

Note that there are other ways to solve this question. You will be given all the points if you have any correct argument.

Problem 9. [10 points]

We implement a "toy example" of the RSA cryptosystem by choosing two prime numbers $p = 23$ and $q = 37$. Assume that we use $(851, 17)$ as the public key in this example. Compute the private decryption key. Show all your calculations.

Answer. $n = p \cdot q = 23 \times 37 = 851$, $e = 17$.

$t = \phi(n) = (p-1)(q-1) = (23-1) \times (37-1) = 792$.

The private key d satisfies $e \cdot d \equiv 1 \pmod{t}$, which means $d = (17)^{-1} \pmod{792}$.

Using the extended Euclidean algorithm.

$$\begin{array}{lccc} \gcd(792, 17) & 792 = 46 \times 17 + 10 & 10 = 792 - 46 \times 17 & (1) \\ \gcd(17, 10) & 17 = 1 \times 10 + 7 & 7 = 17 - 1 \times 10 & (2) \\ \gcd(10, 7) & 10 = 1 \times 7 + 3 & 3 = 10 - 1 \times 7 & (3) \\ \gcd(7, 3) & 7 = 2 \times 3 + 1 & 1 = 7 - 2 \times 3 & (4) \end{array}$$

Substituting:

$$\begin{aligned} 1 &= 7 - 2 \times 3 && \text{(By (4))} \\ &= 7 - 2 \times (10 - 1 \times 7) && \text{(By (3))} \\ &= (-2) \times 10 + 3 \times 7 \\ &= (-2) \times 10 + 3 \times (17 - 1 \times 10) && \text{(By (2))} \\ &= 3 \times 17 + (-5) \times 10 \\ &= 3 \times 17 + (-5) \times (792 - 46 \times 17) && \text{(By (1))} \\ &= (-5) \times 792 + 233 \times 17 \end{aligned}$$

Therefore, the private key $d = 233$.

Remark: d must be a positive number. Considering that we are using $C^d \bmod n$ to compute the original message M, it is necessary to compute a positive multiplicative inverse.

Grading Scheme:

- 2 points for showing that $t = \phi(n) = (p-1)(q-1) = (23-1) \times (37-1) = 792$.
- 2 points for finding the initial linear combination in the extended Euclidean algorithm.
- 1 point for each step of the extended Euclidean algorithm.
- All 5 points for executing the Euclidean algorithm correctly and finding the final answer $d = 233$.

Problem 10. [10 points]

Find all the solutions in the set of integers to the following system of equations:

$$\begin{aligned} x &\equiv 2 \pmod 3 \\ x &\equiv 4 \pmod 5 \\ x &\equiv 3 \pmod 7 \end{aligned}$$

Show all your calculations. You may use any facts and theorems taught in class.

Answer.

We can use the *Chinese remainder theorem* (CRT) to solve this problem.

$$m = \prod_{i=1}^{3} m_i = 3 \times 5 \times 7 = 105$$

$$\begin{aligned} M_1 &= \frac{m}{m_1} = \frac{105}{3} = 35 \\ M_2 &= \frac{m}{m_2} = \frac{105}{5} = 21 \\ M_3 &= \frac{m}{m_3} = \frac{105}{7} = 15 \end{aligned}$$

$$\begin{aligned} y_1 &\equiv (M_1)^{-1} \equiv (35)^{-1} \equiv 2^{-1} \equiv 2 \pmod 3 \\ y_2 &\equiv (M_2)^{-1} \equiv (21)^{-1} \equiv 1^{-1} \equiv 1 \pmod 5 \\ y_3 &\equiv (M_3)^{-1} \equiv (15)^{-1} \equiv 1^{-1} \equiv 1 \pmod 7 \end{aligned}$$

$$\begin{aligned} x &\equiv \sum_{i=1}^{3} a_i M_i y_i \pmod m \\ &\equiv 2 \times 35 \times 2 + 4 \times 21 \times 1 + 3 \times 15 \times 1 \pmod{105} \\ &\equiv 140 + 84 + 45 \pmod{105} \\ &\equiv 269 \pmod{105} \\ &\equiv 59 \pmod{105} \end{aligned}$$

Therefore, the solutions are $x = 105n + 59$ ($n \in \mathbb{Z}$).

Grading Scheme:

The final answer is worth 6 points. 4 points for the procedure.

For the solution with CRT:

- 2 points for showing $M_1 = \frac{M}{m_1} = 35, M_2 = \frac{M}{m_2} = 21, M_3 = \frac{M}{m_3} = 15$.
- 2 points for showing that $y_1 = M_1^{-1} \pmod{m_1} = 2, y_2 = M_2^{-1} \pmod{m_2} = 1, y_3 = M_3^{-1} \pmod{m_3} = 1$.

For the solution with enumeration:

- 4 points for writing out every number satisfying each equation within 105 and declaring that 59 is the common solution.

For other solutions derived from the equations directly:

- 2 points if correctly merging any 2 equations
- 2 points for merging all equations into one

Problem 11. [15 points]

Let $p \geq 7$ be a prime. Prove that the number with decimal representation

$$\underbrace{1111\cdots1}_{\text{p-1 times}}$$

is divisible by p.

Note: For example, in the special case $p = 7$, note that 111111 is divisible by 7, since $111111 = 7 \times 15873$.

Hint 1: Express the number as the sum of powers of 10.

Hint 2: You may find Fermat's Little Theorem helpful.

Answer.

Call the given number S. We want to prove that $p \mid S$. Following Hint 1, note that S can be written as:

$$S = 10^0 + 10^1 \cdots + 10^{p-2}.$$

This is the sum of the first $p-1$ terms of a geometric series, for which we can apply the formula:

$$r^0 + r_1 \cdots + r^{n-1} = \frac{r^n - 1}{r - 1},$$

where $r \neq 1$. Substituting $r := 10, n := p - 1$, we obtain:

$$\begin{aligned} S &= 10^0 + 10^1 \ldots 10^{p-2} \\ &= \frac{10^{p-1} - 1}{10 - 1} \\ &= \frac{10^{p-1} - 1}{9}. \end{aligned}$$

Rearranging, we get that:

$$9S = 10^{p-1} - 1.$$

Taking this equation mod p, we get:

$$9S \equiv 10^{p-1} - 1 \pmod{p}. \qquad (*)$$

Since p is a prime greater than or equal to 7, it follows that p and 10 have no common factors. Following Hint 2, we note that Fermat's Little Theorem is applicable here, and we can use it to obtain

$$10^{p-1} \equiv 1 \pmod{p}. \qquad (**)$$

Substituting from (**??**) into (**??**), we get:

$$\begin{aligned} 9S &\equiv 10^{p-1} - 1 \\ &\equiv 1 - 1 \\ &\equiv 0 \pmod{p}, \end{aligned}$$

which implies that

$$p \mid 9S.$$

Since p is a prime and $p \mid 9S$, it follows that either $p \mid 9$ or $p \mid S$, but since $9 = 3 \cdot 3$ and $p \geq 7$, it cannot be the case that $p \mid 9$. Therefore, we conclude that $p \mid S$.

Grading.

- 4 points for using the formula for geometric sums.
- 6 points for using Fermat's little theorem in a manner that simplifies the problem.
- 5 points for arriving at the conclusion, deducting 2 points if the solution does not provide justification for getting rid of the factor of 9.

Problem 12. [10 points]

(a) Let a and m be positive integers. Prove that no positive integer smaller than $\gcd(a, m)$ can be expressed as an integer linear combination of a and m.

(b) Let a and m be positive integers. Use the fact given in part (a) to prove that if $\gcd(a, m) > 1$, then a does not have a multiplicative inverse modulo m.

Answer.

(a) Let $d = \gcd(a, m)$. Let y be a positive integer that is an integer linear combination of a and m, that is, $y = ak + mk'$, for some integers k and k'. By definition of gcd, we have $d \mid a$ and $d \mid m$. It follows that $d \mid ak + mk'$. Since $y = ak + mk'$ is a positive integer, it cannot be smaller than d.

(b) Assume that a has a multiplicative inverse modulo m. Let us denote it a^{-1}. By definition, $aa^{-1} \equiv 1 \pmod{m}$. It follows that $aa^{-1} - mk = 1$ for some integer k. In other words, 1 can be obtained as an integer linear combination of a and m. But 1 is smaller than $\gcd(a, m)$, which contradicts part (a). This completes the proof.

Grading.

- Each part carries 5 points (partial credit will be given if the proof has minor errors).